Hands-On Lab

Creating A Linux Virtual Machine

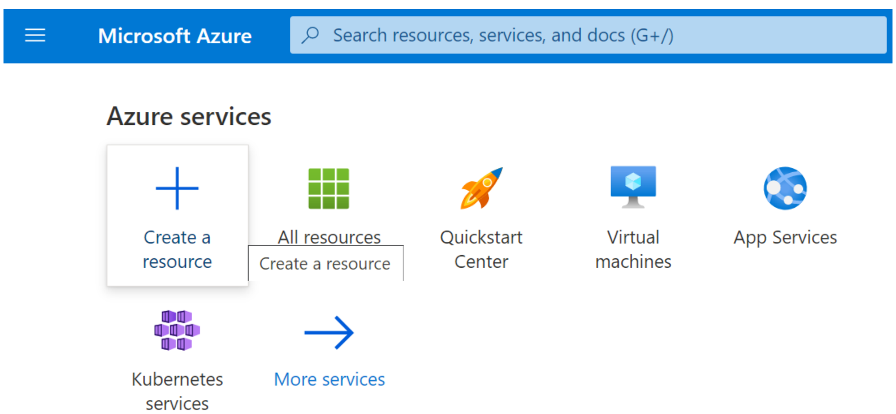
ScholarHat

Azure Linux Virtual Machine (VM)

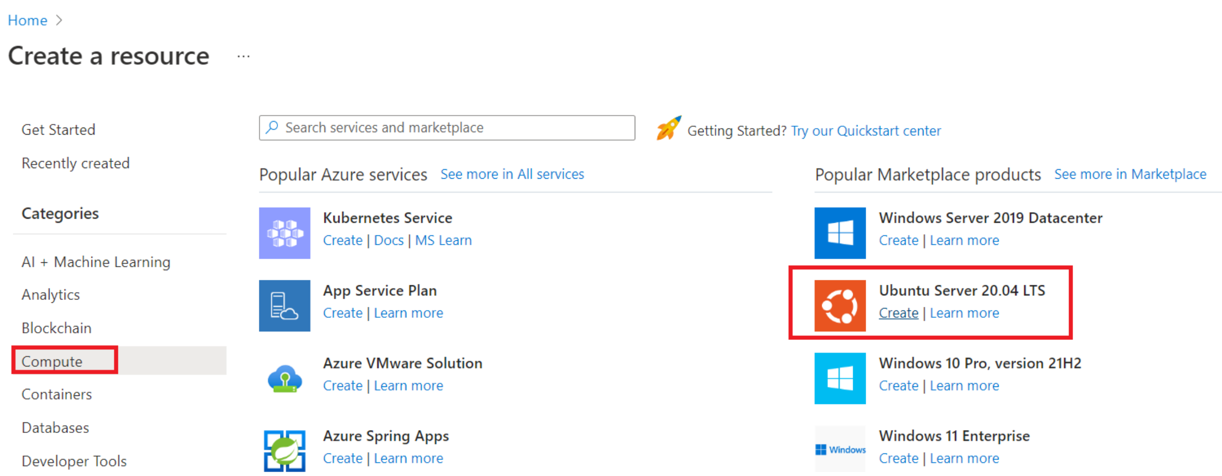
Virtual Machines create a software environment that mimics computer hardware. An operating system can then be installed into this environment, this OS is called “guest OS”.

The operating system installed on the physical computer is called “host OS”. A Linux Virtual Machine is simply a virtual machine (VM) that is using Linux as the guest OS. A Linux virtual machine can also exist on a host server that’s running a non-Linux OS like Windows.

- **Step 1: Open the Azure management portal and log in to https://portal.azure.com**
- **Step 2: Click on “Create a Resource”.**

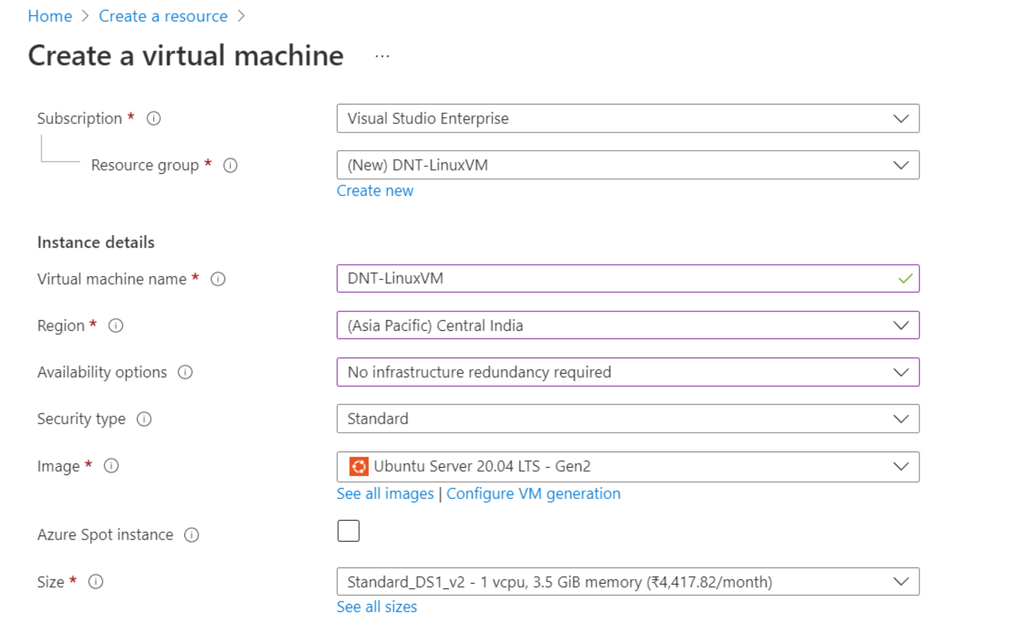


- **Step 3: Click on “compute” and then, hit the create option below the “Ubuntu Server 20.04 LTS”.**

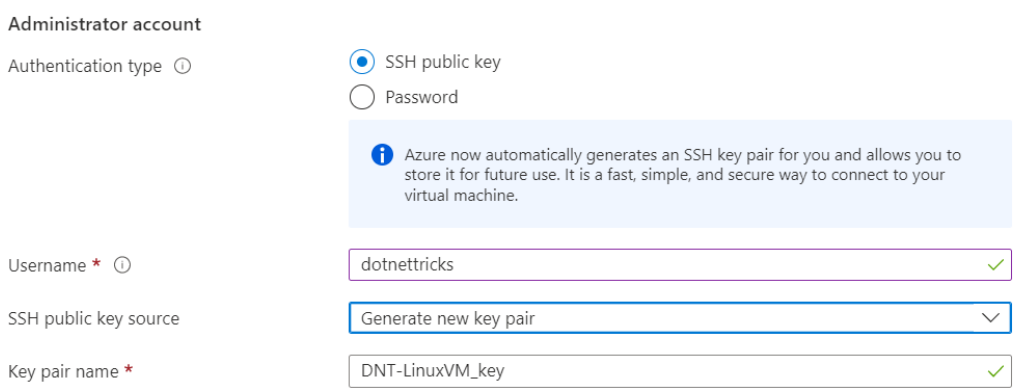


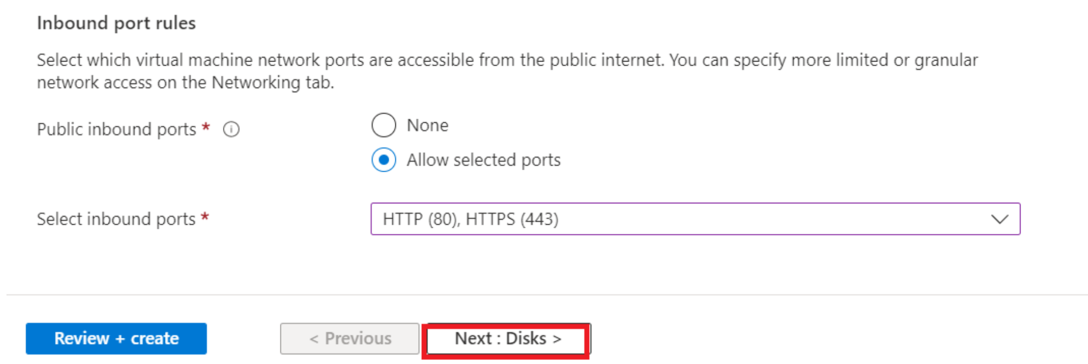
- **Step 4: After clicking create option, Fill in the required details. You have to keep some options as it is but according to your need, you can change some of them.**

1. **Subscription:** Choose a suitable subscription.
2. **Resource group:** Resource group is a container that holds related resources for an Azure solution. You can keep the resource group as it is or you can also create one by clicking “Create New”.
3. **Virtual Machine name:** Mention a name for your Linux virtual machine.
4. **Region:** Here you are going to create all the resources associated with your virtual machine.
5. **Availability Zone:** Select No infrastructure redundancy required.
6. **Security Type:** You can keep it as its default value i.e., standard.
7. **Image:** Select Ubuntu Server 20.04 LTS – Gen2.
8. **Size:** You need to mention size as per your requirements.



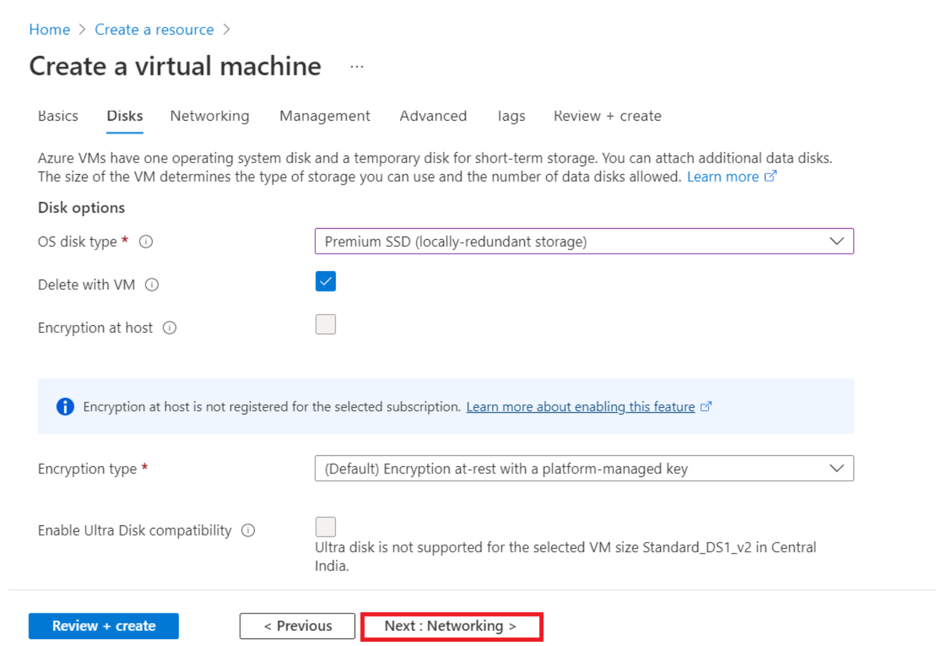
- **Step 5: Now, In the Administrator Account section, Select the SSH public key. In the Inbound port rules section, Select HTTP(80), and HTTPS(443).**
  1. Port 80 is the port number assigned to the commonly used internet communication protocol, Hypertext Transfer Protocol (HTTP). It is the default network port used to send and receive unencrypted web pages.
  2. Port 443 is a virtual port that computers use to divert network traffic. Any web search you make, your computer connects with a server that hosts that information and fetches it to you.





- **Step 6: Now, Hit the "Next: Disks>" button at the bottom for further configurations.**

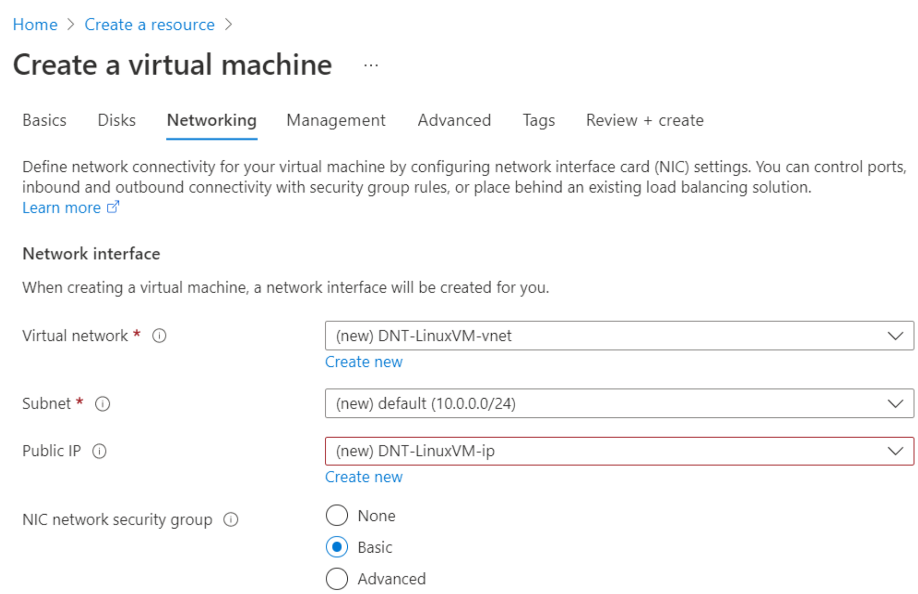
- **Step 7: Firstly, you need to select the OS disk type. Here, we selected Premium SSD. We are provided with three options – Premium SSD, Standard SSD and standard HDD.**
    1. Premium SSDs deliver high performance and low-latency disk support for VMs with high IOPS.
    2. Standard SSDs are optimized for workloads that need consistent performance at lower IOPS.
    3. At last, Standard HDDs delivers reliable, low-cost disk support for VMs running latency-tolerant workloads.

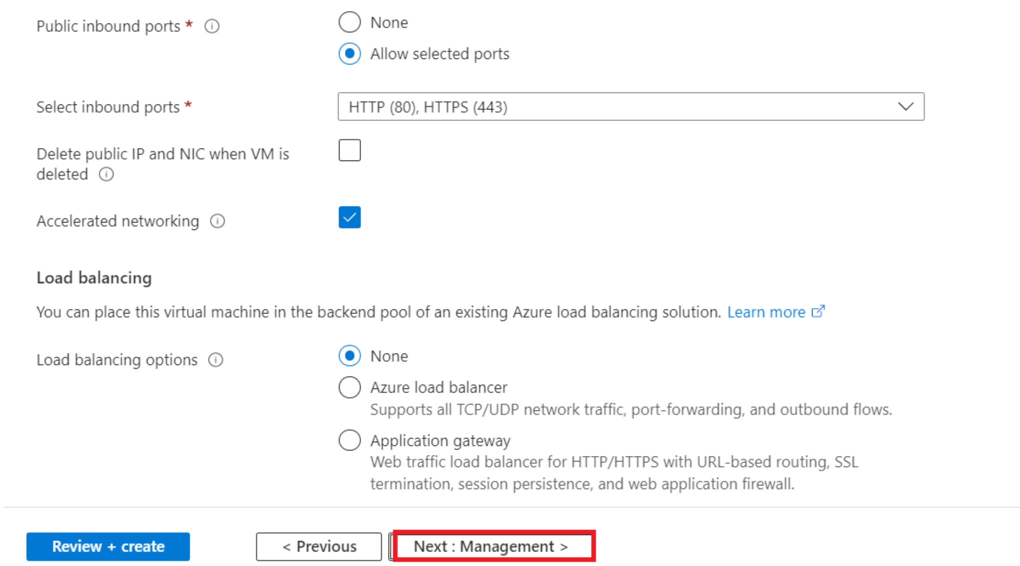


- **Step 8: After completing the details in the disks section, You have to click "Next: Networking >" at the bottom to jump on the next.**

- **Step 9:** **Fill these credentials in the Networking section.**

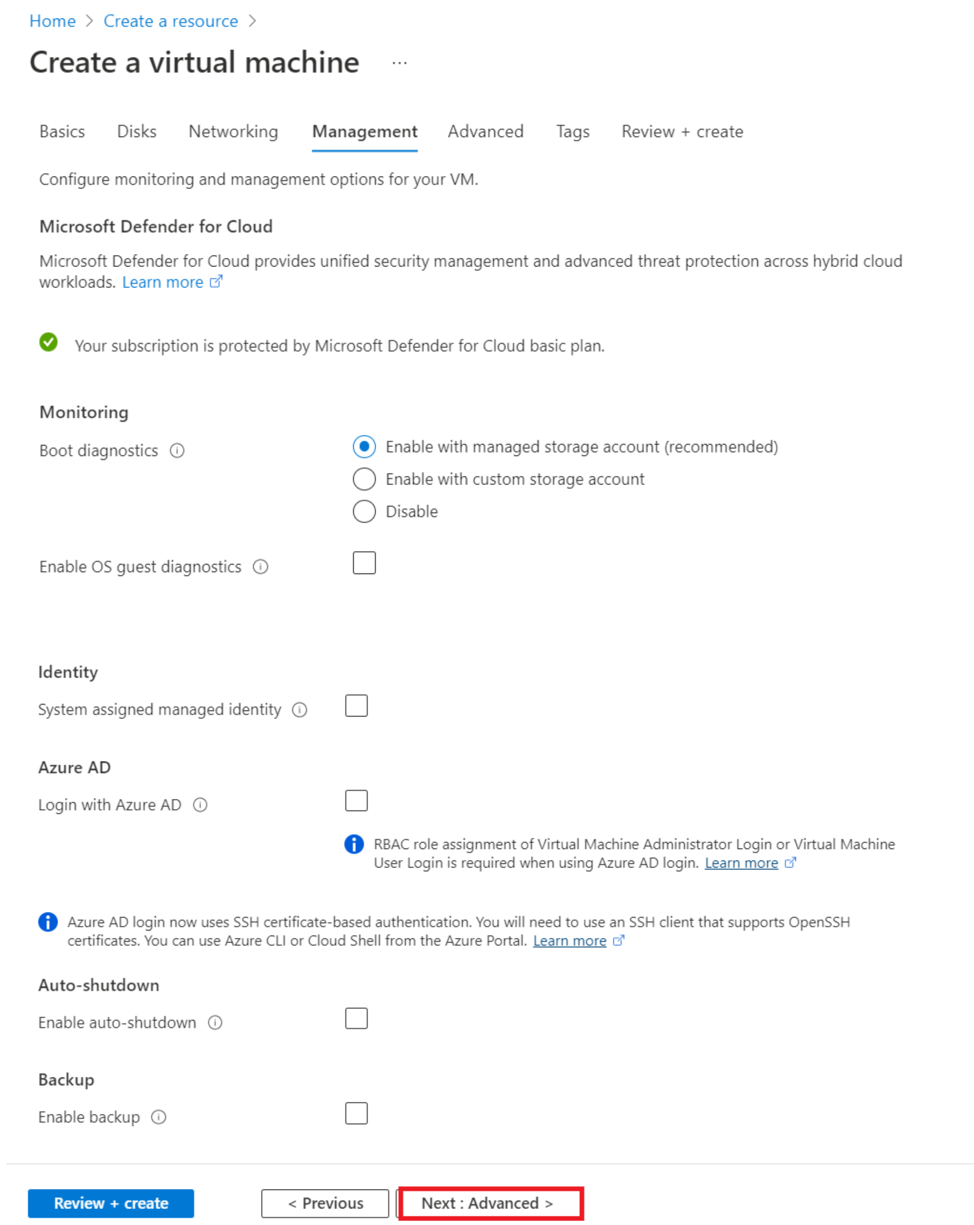
1. **Virtual Network:** Virtual network will be created by default, and you can also create a new one.
2. **Subnet:** You can keep this option as it is.
3. **NIC (Network Security Group):** You can keep this as default i.e., Basic.
4. **Public inbound ports (IP):** Although it will be already selected if not, you can manually select "Allow selected ports".
5. **Select inbound ports (IP):** Choose the option SSH as you've done in the Basics tab.
6. **Load Balancing:** You can keep it as None.



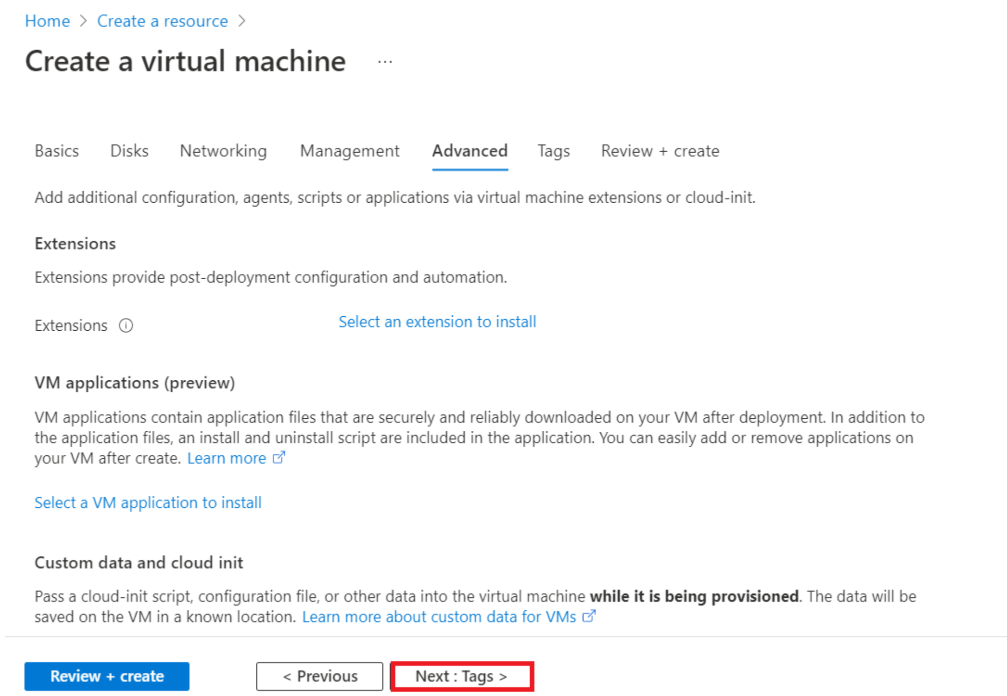


- **Step 10: Now, head towards the next step by clicking on "Next: Management >".**

1. Boot Diagnostics logs startup events and is helpful to trace in case VM does not start by seeing logs.
2. OS guest diagnostics are generally written to an external storage account and are helpful in case of creating alerts.
3. To reduce bills, the Auto shut down feature shut down the VM during the off-hours.
4. We can assign a backup policy that defines the frequency of backups and their retention period.



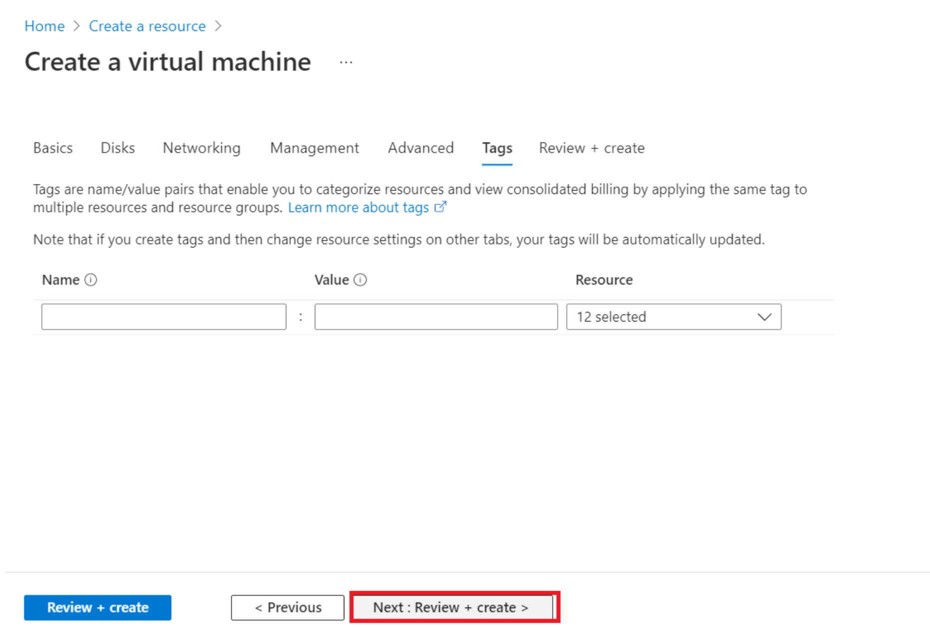
- **Step 11: Jumping onto the next that is Advanced. Here, you bring the VM to the desired state after initialization and Install some extensions which can run scripts on startup. You can keep these options as it is.**



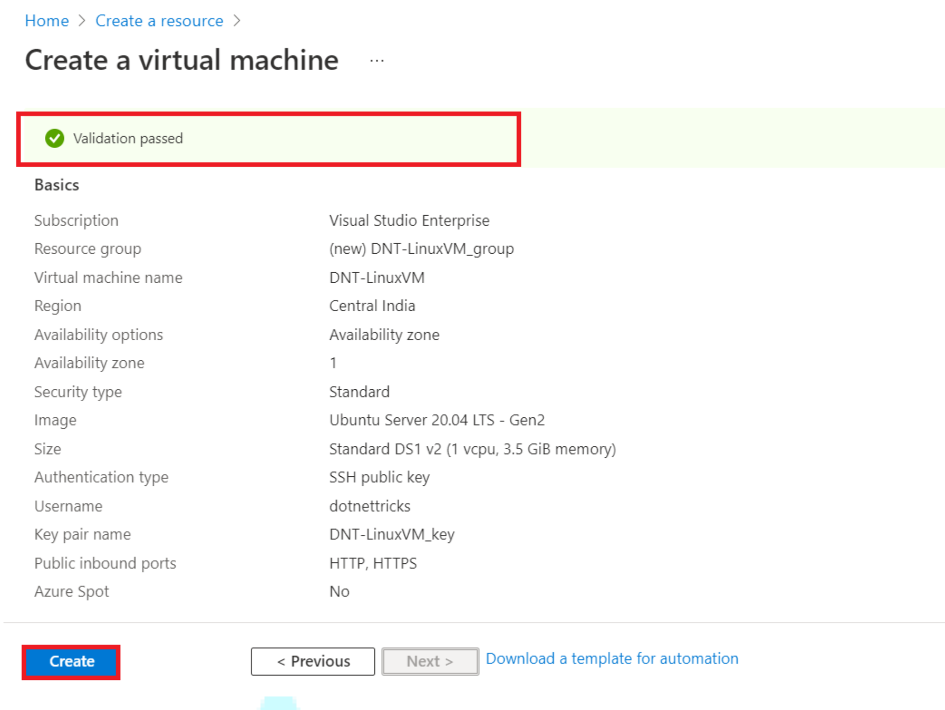
- **Step 12: Now, the next section is Tags. Keep it as it is.**

  Tags are user-defined key/value pairs that can be placed directly on a resource or a resource group. Tags may be placed on a resource at the time of creation or added to an existing resource. We can also see consolidated billings by applying the same tag to multiple resources.

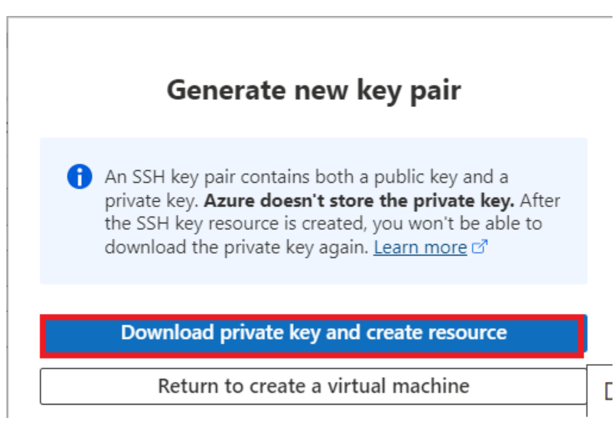
  Afterwards, click on the "Review + Create" button.



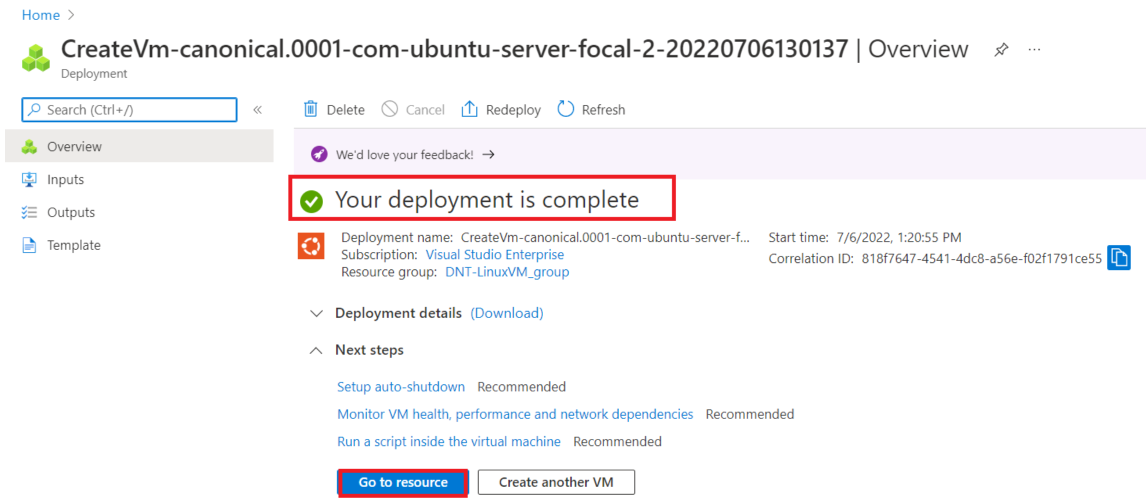
- **Step 13: Azure will authenticate the details you've filled so far and will show a message as "Validation Passed".**
  Afterwards, click on the "Create" button.



- **Step 14: Now, a popup appears to generate the new key pair. Click "Download private key and create resource" and a file with the .pem extension will be downloaded at your desired location.**

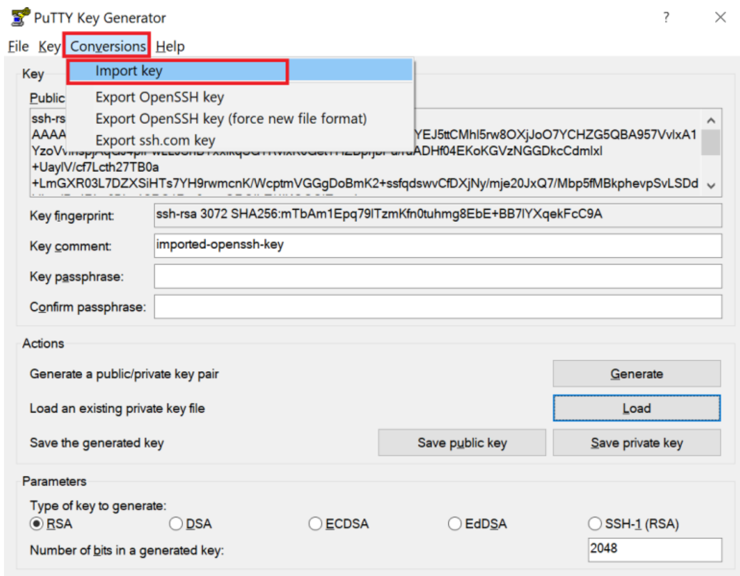


- **Step 14: After clicking the CREATE button. A message will be displayed – "Your deployment is completed". Then click on "Go to resource".**

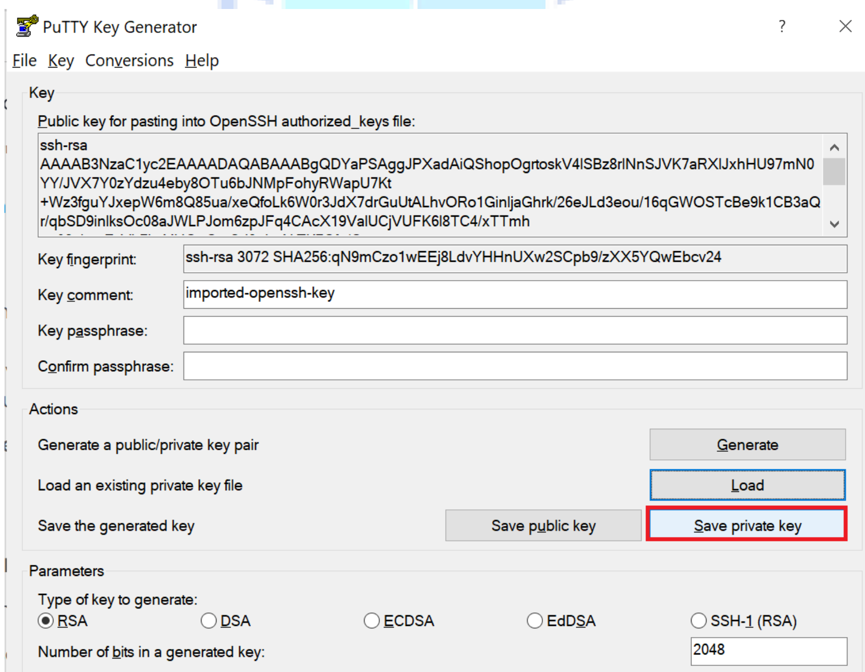


- **Step 15: Now, You have to connect to a Linux Virtual Machine using PuTTY.**
  - **Download PuTTY from the official website - https://www.putty.org/**
  - **Download PuTTYgen - https://www.puttygen.com/download.php?val=4**

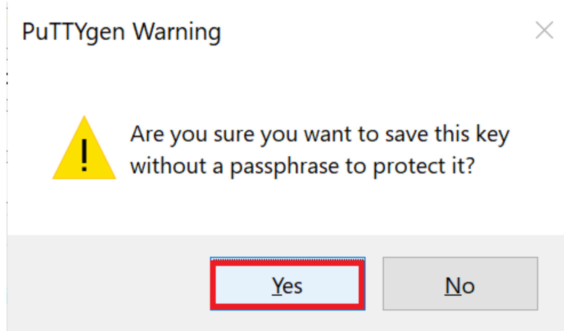
- **Step 16: Open PuTTYgen first, Go to the Conversions tab and then, Click on the import key.**



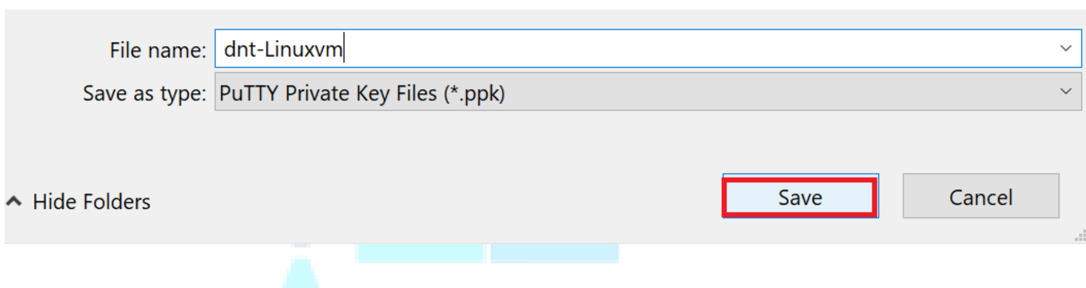
- **Step 16: Upload the .pem file that you have downloaded in Step 14. Click on "Save private key".**



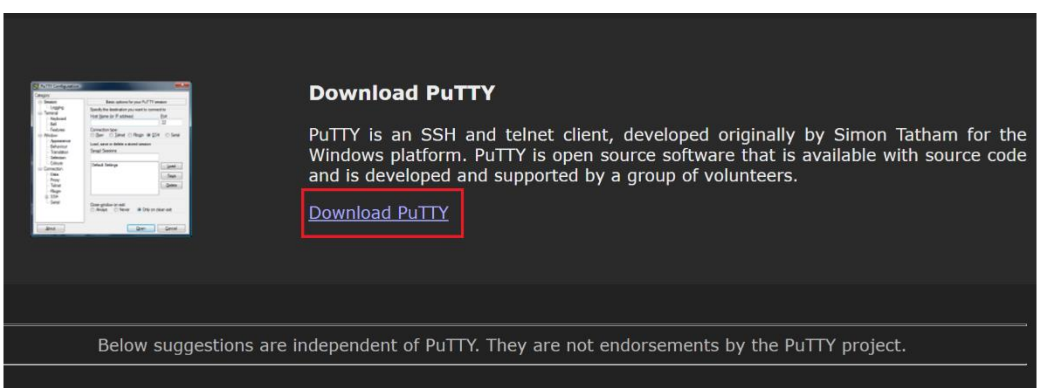
- **Step 16: A popup appears for the confirmation that you're sure you want to make this key without a passphrase to protect it. Click on YES.**

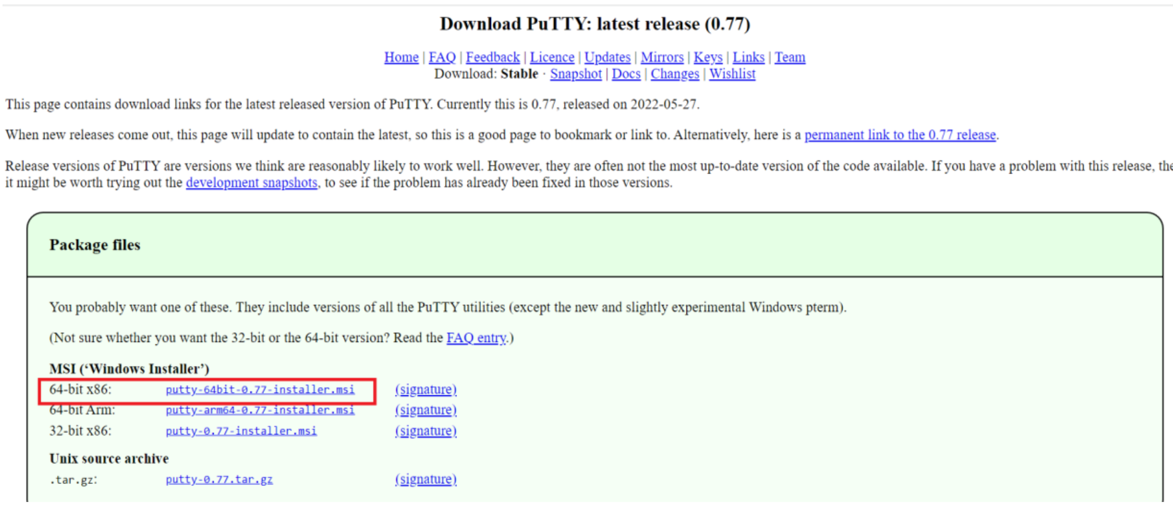


- **Step 16: After clicking on Yes, you need to provide a name for the .ppk file. Click on the "Save" button.**



- **Step 16: Now, You have to download the PuTTY from the link provided above. This page contains the download links to the latest version of PuTTY. Download it by clicking the "64-bit x86" link. Select the desired location for the downloaded file.**





Download PuTTY: latest release (0.77)

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This page contains download links for the latest released version of PuTTY. Currently this is 0.77, released on 2022-05-27.

When new releases come out, this page will update to contain the latest, so this is a good page to bookmark or link to. Alternatively, here is a permanent link to the 0.77 release.

Release versions of PuTTY are versions we think are reasonably likely to work well. However, they are often not the most up-to-date version of the code available. If you have a problem with this release, the it might be worth trying out the development snapshots, to see if the problem has already been fixed in those versions.

Package files

You probably want one of these. They include versions of all the PuTTY utilities (except the new and slightly experimental Windows pterm).

(Not sure whether you want the 32-bit or the 64-bit version? Read the FAQ entry.)

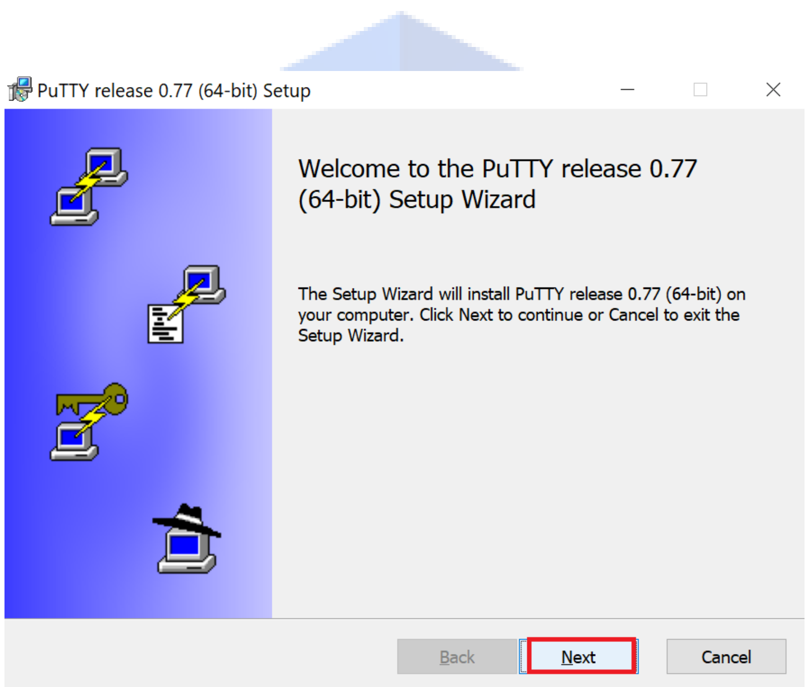
MSI ('Windows Installer')

64-bit x86:	putty-64bit-0.77-installer.msi	(signature)
64-bit Arm:	putty-arm64-0.77-installer.msi	(signature)
32-bit x86:	putty-0.77-installer.msi	(signature)

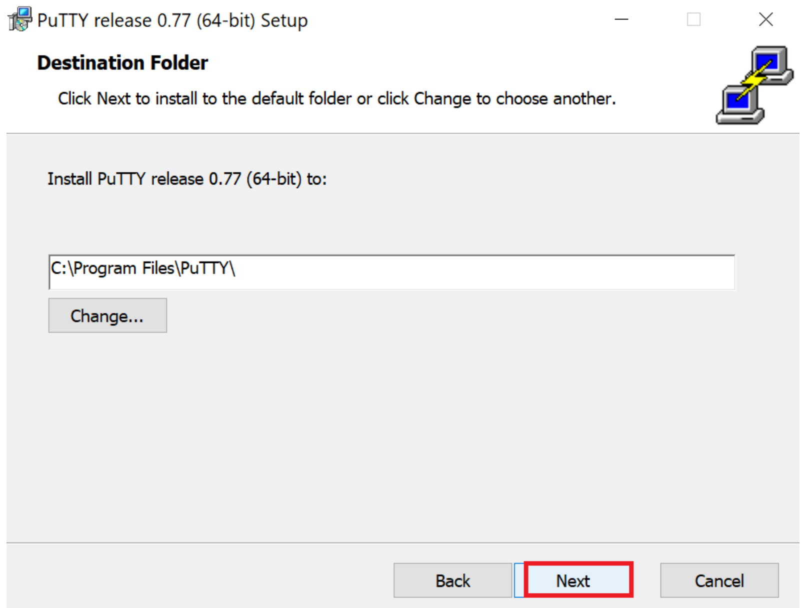
Unix source archive

.tar.gz:	putty-0.77.tar.gz	(signature)

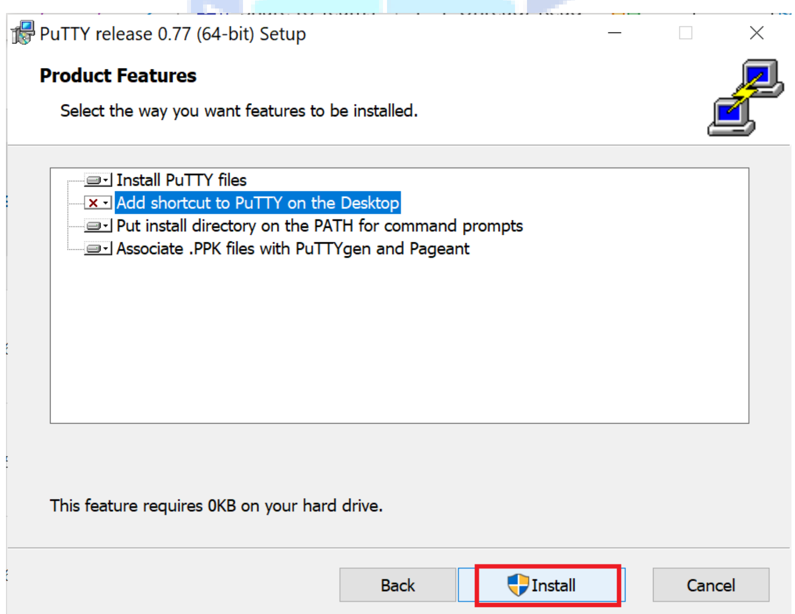
- **Step 17: Open and Run the downloaded PuTTY Windows installer package. As soon as you run the file, the Setup Wizard appears. Click on "Next".**



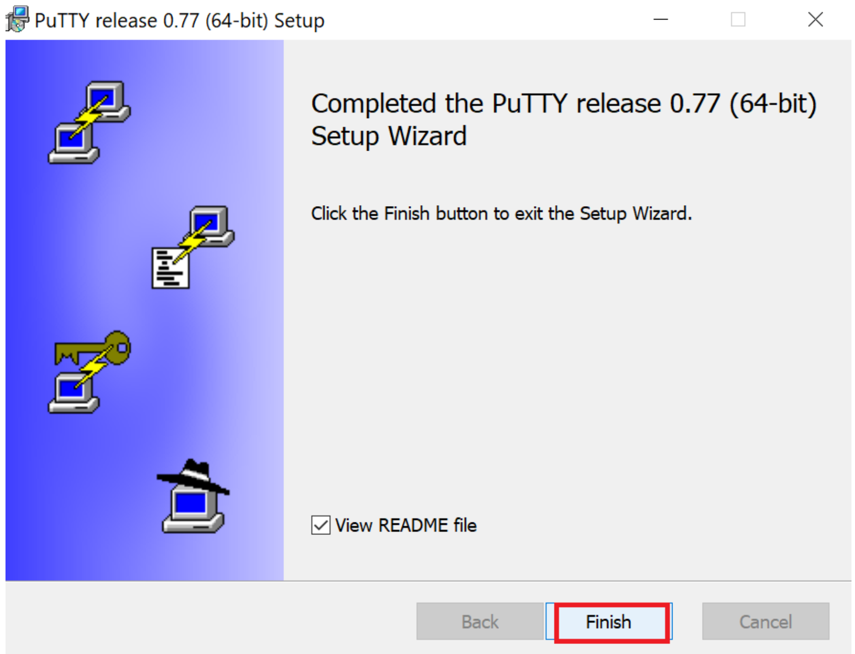
- **Step 18: Select the Destination Folder. Then, Click on "Next".**



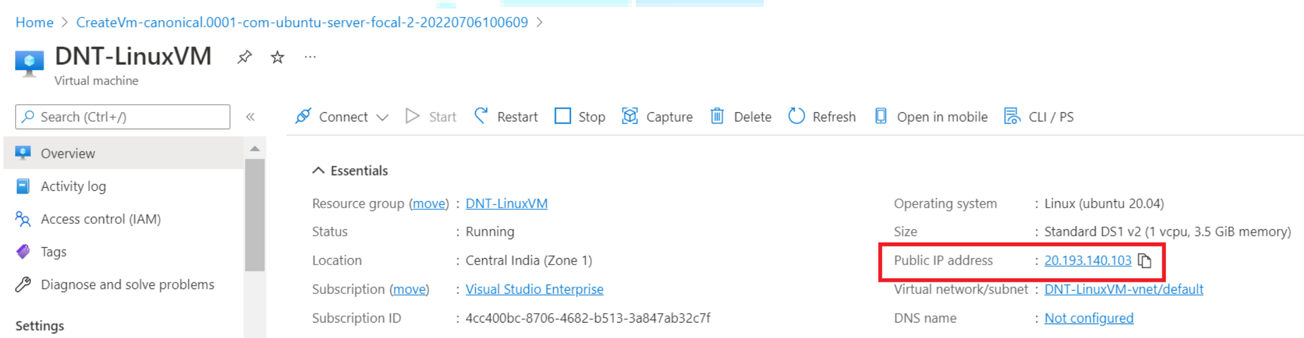
- **Step 19: Select the way you want the features to be installed. Just click on "Install" to start the installation process.**



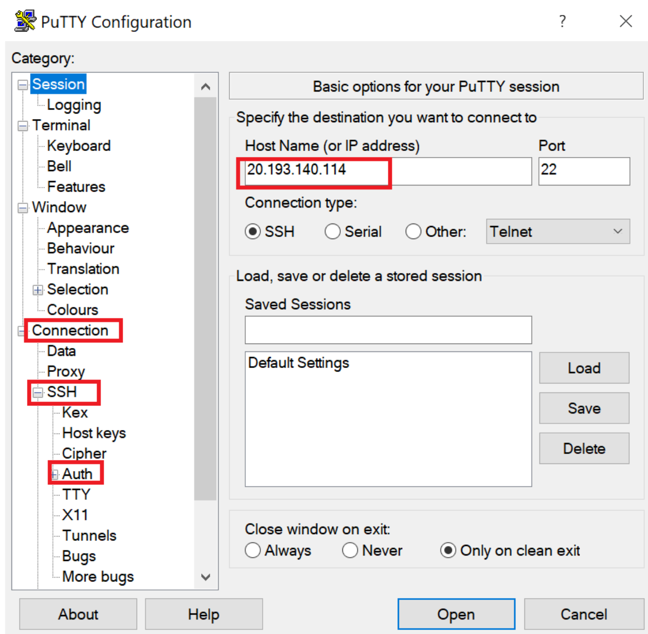
- **Step 19: It will start installing and will take a few seconds. Then Click on the "Finish" button and You are good to go!**



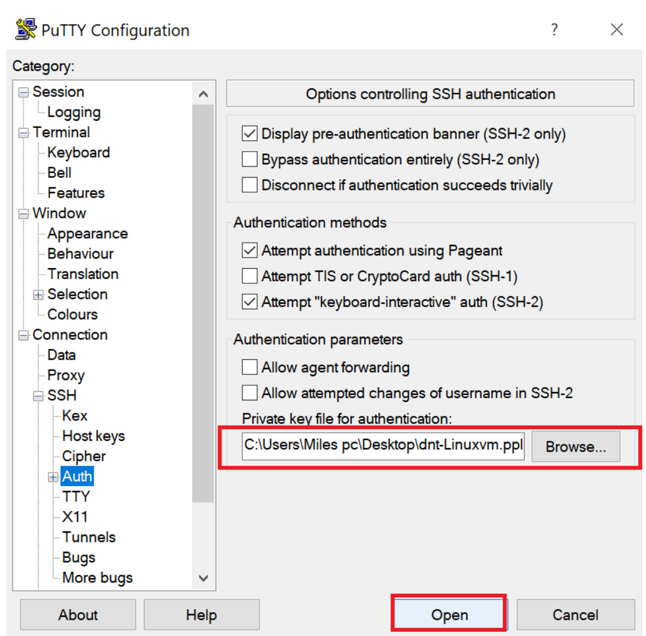
- **Step 20: Now, again visit the Azure portal where you have created your Linux Virtual Machine. You'll see the public IP Address, Just Copy that address.**



- **Step 21: Go to Start Menu, and search and Open the PuTTY app. Paste the copied IP address as shown in the below image. Before Selecting "Open", Click Connection > SSH > Auth tab.**



- **Step 21: The below screen appears. Browse and Select your PuTTY Private Key (.ppk) file which you've saved from the PuTTYgen. Then, Go to the Session tab and Hit the "Open" button.**



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- **Step 21: You have successfully made an SSH connection into a Linux virtual machine.**

Contact Us

You can always reach out to us if you need help with this practice hands-on lab. For getting help email us at hello@scholarhat.com or connect at the **#support** channel on Discord.